

Solving the Model Gap

Why does cheaper, better silicon not win by itself?

**AMD-Sourcing the Engineering Cycles to Close NVIDIA's Physical-AI
Lead -- A proposal**

1 · The Paradox We Keep Hitting in Sales

We Win the Spec Sheet and Lose the Deal

The customer never disputes the economics — he refuses because the decision was made above the silicon.

AMD hardware wins on economics

- ~**50% lower capex** and ~**30% lower opex** vs the incumbent (on the right criteria)
- Competitive silicon: MI300X **5.3 TB/s** HBM3, Ryzen AI edge, Kria deterministic control

On price-performance, we may have/reach a real, quantifiable advantage.

...yet the application developer still says "no"

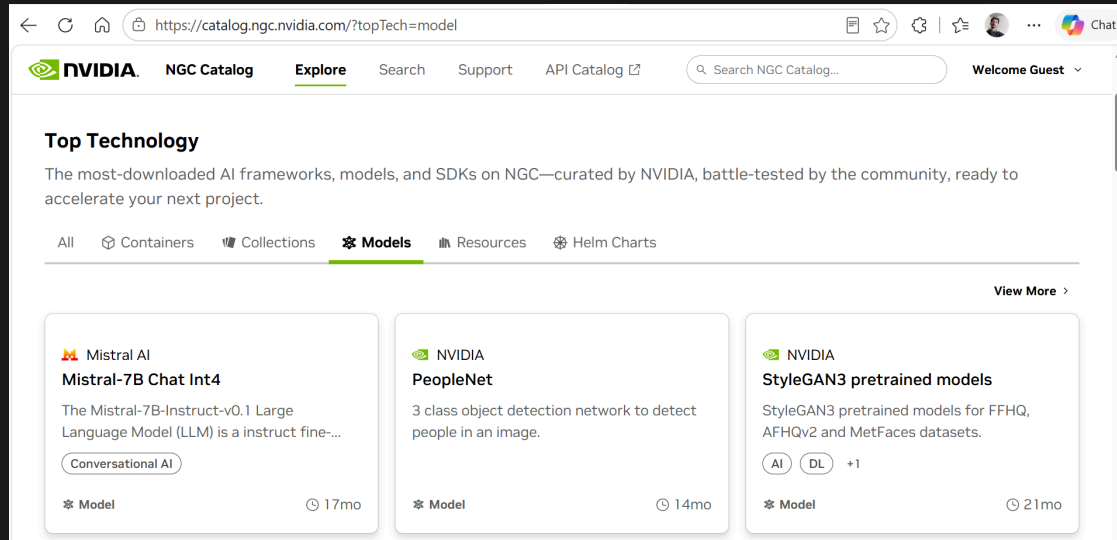
- His product is already built **around NVIDIA's models and pipelines**
- Switching means **re-engineering the whole stack** — a cost *he* has to pay
- The better chip **never gets a fair evaluation**

We lose the deal **before** hardware economics enter the conversation.

The barrier is not the silicon — it's the ecosystem the customer has already built on top of NVIDIA.

2 · NVIDIA's Moat Is Software, Not the Chip

A Decade of Models Turned Into a Vertically Integrated Stack



NGC Catalog — years of curated, prepackaged, ready-to-run models

The lock-in is measurable

- **~80% ecosystem, ~20% hardware** — NVIDIA's own default status
- **2M+ robotics developers, 4M CUDA developers, 7+ year head start**
- **Isaac Sim** leads robotics job postings — **50% more** than the closest competitor
- **1,228 VLA papers** at ICLR 2026 → **zero** mention ROCm

Models → TensorRT → Isaac ROS → cuRobo → Isaac Sim → Cosmos — where each layer locks in the next — and locks AMD out.

3 · The Switching Cost Is the Real Barrier

The Customer Would Take Cheaper Hardware — but He Pays the Switching Bill

Every engineering cycle spent re-hosting on AMD is capex the customer must justify before he saves a cent.

Why the customer stays put

- Years of infra, pipelines, and applications built on NVIDIA models
- Rebuilding fleet/deployment infra: **6-18 engineer-months per deployment**
- **5-7 deployment artifacts per model** to re-target hardware
- Even hardware-competitive rivals face a **12-24 month** software catch-up

The burden lands hardest where we want to win

Small / mid-size robotics companies

every engineering cycle is capex they cannot spare



They keep buying the more expensive chip — because **switching costs more than it saves**

This is the soft-power gap — AMD is absent from every on-ramp above the compute layer, and it shows up directly in sales.

4 · The Internal Problem: ROI Can't Be Justified Overnight

The Gap Is Known and Winnable — But ROI can not be defended

How AMD is structured

- Many business units — each funded by its **own sales / engineering output**
- Every unit must **defend the returns** on what it takes on

Why no single unit picks this up

- Model training is **iterative and continual** — not a one-shot deliverable
- A **multi-year ecosystem gap** cannot be closed overnight
- ROI in a fixed window is **hard to measure and harder to justify**

The Deadlock

Gap is **large** & the payoff is **long-horizon**



No BU can **book near-term ROI**



So **nobody starts** — and the gap widens

A multi-year, iterative gap cannot be booked as near-term ROI — so every business unit is rational to pass.

5 · The Insight: The Target Is Already Known

When the models and target numbers are published, closing the gap becomes bounded, measurable work.

Unlike open-ended research, we are not searching

- The **models are named** and mostly **open** (NGC is a public map)
- The **target performance is published** — we know exactly what "done" looks like
- This is a **replication problem**, not an invention problem
- The **replication problem**, can be solved at scale

Producing a replica of a known target is **bounded, measurable work** — the ROI objection shrinks.

Only two pieces are missing

Compute — we already have it

AMD-wide clusters (ALOLA & others); PAVS can broker access

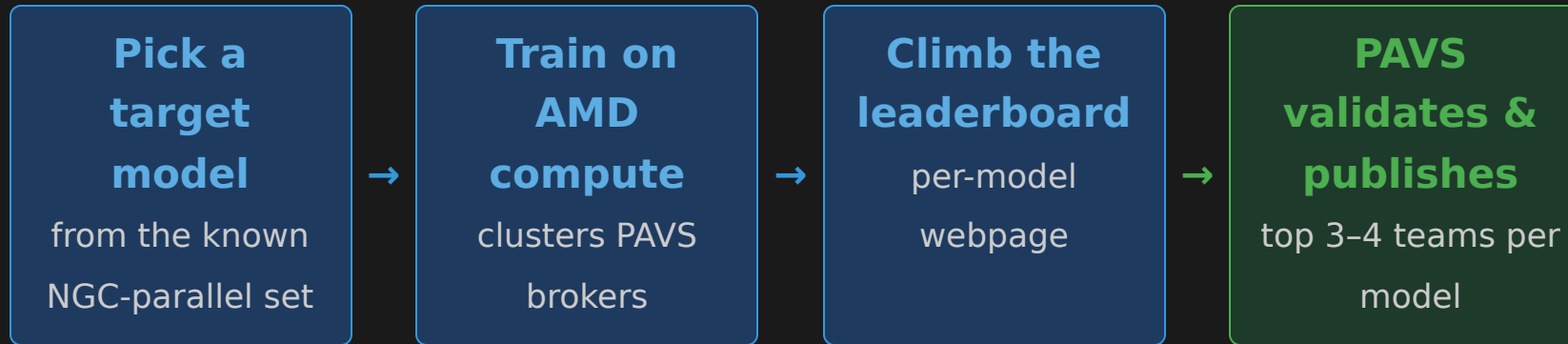
Data — the real problem to solve

synthetic generation · **pooled + curated open data** · **semi-supervised for the task** · **pseudo-labels from large OSS models** · & more

Known target + available compute → the task collapses to **solving the data problem at scale.**

6 · The Proposal: An AMD-Wide Model Training Challenge

Convert one multi-year bill into a cohort of motivated engineers, pointed at a scored target.



The mechanics

- A **leaderboard per model** (grouped: perception · manipulation · VLA · SLAM ...)
- A **webpage** tracking the top teams and their models
- Each **new #1 notifies every participant** — winner + solution shared — keeping the field competitive
- **Top 3-4 teams** get to **publish their work** under AMD
- PAVS **reproduces and certifies** the model +

Why a challenge, not a hire

- Taps **latent enthusiast engineering** across all of AMD
- Converts a **multi-year gap** into **many parallel short sprints**
- No single BU carries the bill — the org **contributes cycles**

7 · Challenges Have Closed Big Capability Gaps Before

We're not guessing this works — a scored, public target has repeatedly closed gaps no single roadmap could.

ImageNet Challenge

- A **public leaderboard** on a known target
- Sparked **AlexNet** and the deep-learning era
- A benchmark did what no single lab's roadmap could

DARPA Grand Challenge

- A **prize + leaderboard** for autonomous driving
- Year 1: nobody finished. Year 2: **multiple teams did**
- Seeded the entire self-driving industry

Kaggle / Netflix Prize

- **Crowdsourced** solutions to a fixed, scored target
- Outperformed in-house teams at a **fraction of the cost**
- Recognition — not salary — drove the effort

"The right challenge stimulates the right engineer." — a scored, public target closes gaps faster than any single funded roadmap.

8 · The Target Set Is Open — and Achievable

The Models Are Open, the Datasets Exist, the Data Scales Cheaply

One AMD alternative per NGC model — the perception + manipulation set Isaac ROS consumes — each with an open replica to train.

One replica per NGC / Isaac ROS model

Isaac ROS model (NGC)	Open replica to train on AMD
Detection — PeopleNet / DetectNet	YOLOv12 · RT-DETR
Segmentation — PeopleSemSegNet	SegFormer · U-Net
Stereo depth — ESS	RAFT-Stereo · HITNet
6-DoF pose — FoundationPose / DOPE	FoundationPose · MegaPose
Visual SLAM — cuVSLAM	ORB-SLAM3
Manipulation / VLA — GR00T / pi0	OpenVLA · SmolVLA

The models are reachable, the data exists

- **OpenVLA** fine-tunes on **1× A100 in <24h**; **SmolVLA** needs **50× less** compute
- **Perception** — COCO · KITTI · nuScenes public benchmarks
- **Manipulation** — Open X-Embodiment (1.4M episodes) · LeRobot (58k datasets)
- Synthetic data is **50-70× cheaper** than real-world capture
- The challenge lets **many teams attack the data problem in parallel**

Each NGC model has a **concrete, scored finish line** — replicate it, hit the target on AMD



Each replica drops into ****AMD-ROS / AARTS**** —

9 • Incentive Design — Everyone Wins

A Challenge Only Works if Every Party Wins by Playing — Here, Three Do

The engineer, PAVS, and AMD each get something they want — which is why the mechanism sustains itself.

Participant (Engineer)

- **Recognition** — publish work, top-team webpage
- A **countable contribution to AMD** for their record
- An **extra purpose & goal** alongside the day job
- Enthusiast hours become **productive output**

Organizer (PAVS)

- Becomes the **customer gateway** to physical AI across AMD
- Reproducing models **builds working ties** across many teams
- A **niche — but the lead player** in that niche
- **Bridges the islands** of physical AI at AMD

AMD (Org)

- The **ecosystem gap closes fast** — target is known
- A growing **pool of curated datasets** — one team's data & models **catalyze the next**
- **Idle compute** (ALOLA & others) gets **popularized**
- A pipeline of **market-ready, AMD-native models**
- Engagement without a **single-BU budget line**

One mechanism, three winners — which is why a crowdsourced challenge is self-sustaining, not a one-off spend.

10 · PAVS as the Customer Gateway to Physical AI

Organizing the Challenge Isn't Overhead — It's How PAVS Earns Its Position

Reproducing and routing every winning model is exactly what makes PAVS the gateway physical AI flows through.

What PAVS commits to

- Send an **org-wide call** for the right technical solutions
- The right solution that reaches us → **we help take it to market**
- **Reproduce, evaluate, and certify** each model + its performance
- Curate the leaderboard and the **model webpage**

What that makes PAVS

The gateway — every physical-AI solution enters through PAVS



The bridge — connecting the islands of physical AI across AMD



The lead player in a well-defined niche

The organizing work is not overhead — it's how PAVS earns the relationships and the position to route physical AI to market on AMD.

11 · An Enormous Market — The Prize Is Worth the Effort

A drop-in AMD replica per NGC model turns a costly migration into a seamless switch — and unlocks the customers already locked in.

\$200B

AMD's own projection for the physical-
AI silicon market by 2035

\$22.2B

robotics startup funding in 2025 — up
69% YoY; the buyers of tomorrow are
forming now

0

of 1,228 VLA papers at ICLR 2026
mention ROCm — the on-ramp is wide
open to claim

**Make the transfer seamless and AMD's price-performance is finally evaluated — the
cheaper stack wins the deals it loses today.**

The Ask

The plan is already staffed — we only need the mandate to start

What we need from leadership

An **org-wide email** endorsing the challenge — the signal that gives the effort **traction**.

Already staffed & committed — **execution is ready**

Compute — grants cluster access (ALOLA & AMD-wide)



Steggink, Matthew
PMTS Software Development Eng. • ML LIBRARIES CAN



Per-model GitHub — creates the per-model repos



Misra, Bharat
Manager Software Development • Embedded Prod Dev



Team to drive the work



Patil, Vijayalaxmi
Sr. Manager Software Development
EMBEDDED PROD DEV

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Vydana, Hari Krishna
SMTS Software System Design Eng.
EMBEDDED PROD DEV
1.09A125



You work with



Sapale, Amol
SMTS Software System Des...
1.09B309



Kokate, Kalyani
Program Manager 2
1.09B310



Gandhi, Vaibhav Rajesh
SMTS Software Developme...
1.09B324



R Kharatmal, Vilas
SMTS Software System Des...
1.09B327

The push

Known target · staffed team · available compute — **all that's missing is the go-ahead.**



together we advance